International Islamic University Chittagong
Center for General Education (CGED)
Semester Ending Examination, Autumn-23
Course Code: URED-3604 (URED-3102 for LLB)
Course Title: Life and Teachings of the Prophet (*SAAS*)

Time: 2:30 Hours **Full Marks: 50**

Answer the following questions
(All questions are of equal value)

SL	Questions	Marks	CLO	Bloom's taxonomy domain
1	"*Hijrah* is the turning point in the history of Islam". Explain the statement discussing the causes, importance and lessons of *Hijrah*	10	3	Create
2	What are the main clauses of the charter of Madinah? How did Prophet (SAAS) establish a society based on peaceful co existence in Madinah ? Explain.	10	2	Understand & Create
3	Analyze the background of battle of Badr assessing the reasons why the Muslims got victory. **or** Analyze the background of battle of *Uhud.* What are the lessons to be learnt from this battle?	10	2	Analyze
4	Explain the background and importance of the Conquest of Makkah **or** "The treaty of Hudaibiah is a clear victory". Explain	10	3	Create
5	Evaluate the Farewell Address of Prophet Muhammad (*SAAS*) as a document of human rights.	10	3	Evaluate

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. Engineering in CSE

Final Exam, Autumn 2023

Course Code: **ECON 3501** — Course Title: **Principle of Economics**

Time: 2 hours 30 minutes — Full Marks: 50

Answer all the questions.

(The figures in the right-hand margin indicate full marks)

Part A

1. a) Define market and market structure. Explain and compare the different types of market structure. — CO1 | An | 5

1. b) The table below shows the market demand schedule and the cost structure. — CO2 | E | 5

Quantity	Price per	Fixed cost (FC)	Variable cost (VC)
20	$18	$200	40
35	$18	$200	70
55	$18	$200	110
75	$18	$200	150
105	$18	$200	210
130	$18	$200	260
160	$18	$200	320
190	$18	$200	380
220	$18	$200	440
250	$18	$200	500

$\frac{TC_2 - TC_1}{Q_2 - Q_1}$

Measure Total cost (TC), Average total cost (ATC) and Marginal cost (MC) for each row.

OR

1. a) Is competition in the market good? Why? Describe the characteristics of a perfectly competitive market. — **CO1** | **An** | 5

1. b) Consider this cost function: $TC = 10+20q+4q^2$ — **CO2** | **E** | 5

Requirement: I. Compute TC, AC and MC when q = 7

II. Find where MC=AC

III. What level of output minimizes AC?

2. a) Suppose, GDP = $5700 — **CO2** | **E** | 5

Net Factor payment from abroad = $18

Capital consumption allowance = $625

Indirect taxes = $475

Social security contribution = $530

Govt. and business transfers to person = $770

Dividends = $135

Personal Tax & Non-Tax payment = $620

Required: Calculate **GNP, NNP, National Income, Personal Income and Personal Disposable income**

2. b) What is Inflation in Economics? What are the causes of Inflation? Discuss briefly. — **CO2** | **An** | 5

Part B

[Answer the questions from the followings]

3.	a)	Discuss monetary policy and fiscal policy. How fiscal policy plays an important role in a developing country like Bangladesh?	CO2	An	5
3.	b)	What do you mean by Subsidy? How does the government give subsidies to the citizens? Describe some benefits of subsidy.	CO1	Un	5
4.	a)	Not only the availability but also ability to utilize the natural resources is important in economic growth. – explain the statement in the case of Bangladesh.	CO1	E	5
4.	b)	Describe some obstacles of economic development.	CO2	An	5
5.	a)	Evaluate the relationship between Technological progress and economic growth in Bangladesh.	CO3	E	5
5.	b)	Explain the difference between growth and development?	CO3	E	5
		OR			
5.	a)	Zakat & Poverty reduction, 8^{th} 5 year plan of Bangladesh, *Unemployment* rate in Bangladesh and Devaluation of Bangladeshi Currency.	CO3	E	10

International Islamic University Chittagong

Department of Computer Science and Engineering

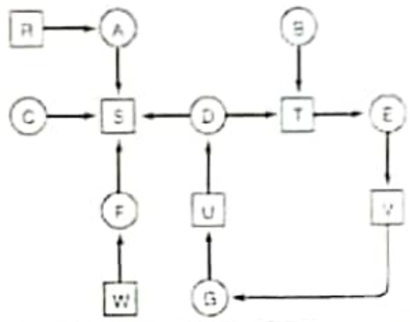
B. Sc. in CSE

Final Exam, Autumn 2023

Course Code: **CSE-3631** | Course Title: **Operating Systems**
Time: 2 hours 30 minutes | Full Marks: 50

Group A

1. a) Define Deadlock. In the resource allocation graph (RAG) shown in the following figure, is there any deadlock, and if so, which processes are involved? In this graph, the square nodes represent resources, and the circular nodes represent processes. — CO2 | Ap | 5



1. b) In a multi-process system, there are three types of resources (X, Y, and Z) with initial availability (8, 6, 4) units, and five processes (P1, P2, P3,P4,P5) with different maximum needs and current allocations. Using the Banker's algorithm, calculate the maximum need matrix, allocation matrix, and available resources at each step to demonstrate how resources are allocated to the processes while maintaining system safety and preventing potential deadlocks. Provide a clear example of the allocation process. — CO2 | Ev | 5

Process	Max/Allocation	Max / Allocation
P1	7 5 3	2 2 2
P2	3 2 2	2 1 0
P3	9 0 2	3 0 2
P4	2 2 2	0 1 1
P5	6 4 2	4 3 1

OR

1. b) Consider the following resource allocation state of a system with four processes, P1, P2, P3, and P4, and five types of resources, RS1, RS2, RS3, RS4, and RS5 where — CO2 | Ev | 5

C :the current allocation matrix,
R: the request matrix.
E :the existing resource vector
A : the available resource vector

$$C = \begin{bmatrix} 0 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 2 & 1 & 0 & 0 & 0 \end{bmatrix} \qquad R = \begin{bmatrix} 1 & 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 2 & 0 & 3 & 1 \\ 0 & 2 & 1 & 1 & 0 \end{bmatrix} \qquad \begin{matrix} E = (24144) \\ A = (01021) \end{matrix}$$

1. Using the deadlock detection algorithm, show that there is a deadlock in the system. Identify the processes that are deadlocked.
2. Explain how the system can recover from the deadlock using recovery through preemption.

2. a) What is paging? Explain the concept of free frames in memory management and how they are used for process allocation. Provide an example of a scenario where you have 16 free frames initially. Two processes, Process A and Process B, request memory allocation. Describe the state of free frames before and after the allocation, and discuss the significance of efficient frame allocation in a multi-process environment. — CO3 | Un | 5

2.	b)	Discuss the impact of page faults on the overall performance of a computer system and explain how page fault handling mechanisms, such as demand paging, help minimize the effects of page faults.	CO3	Un	5
		OR			
2.	b)	In a computer system, the cache has a Cache Hit Time (H) of 1 nanosecond (ns), a Cache Miss Penalty Time (M) of 10 milliseconds (ms), a Cache Hit Rate (Hr) of 90%, and a Cache Miss Rate (Mr) of 10%. Calculate the Effective Access Time (EAT) for this cache configuration. How does EAT provide insight into the performance of the cache in terms of average memory access time?	CO3	Un	5
		Group B			
3.	a)	Explain the LRU (Least Recently Used) page replacement algorithm in the context of memory management. Using a provided sequence of page references and a memory capacity of 4 pages, demonstrate how the LRU algorithm works to manage pages in memory. Calculate the page fault rate for the given sequence. Discuss the advantages and limitations of the LRU algorithm in comparison to other page replacement algorithms. you have the following page reference string: 2 3 2 1 5 2 4 5 3 2	CO3	An	5
3.	b)	Explain the steps involved in handling a page fault in a virtual memory system. Discuss the key responsibilities of the operating system during this process, including checking page validity, locating the required page, selecting victim pages (if necessary), and updating the page table. Provide an example scenario to illustrate the page fault handling process explain with proper diagram.	CO3	Ap	5
		OR			
3	b)	Illustrate with an example how the Enhanced Second-Chance Page Replacement Algorithm works.	CO3	Ap	5
4.	a)	Explain the different file access methods used in operating systems and file systems. Compare and contrast the characteristics of sequential access, direct access, and indexed access methods. Provide examples of scenarios where each access method is most suitable and discuss the advantages and limitations of each in terms of data retrieval and storage.	CO3	Ap	5
4.	b)	Explain the working procedure of Hashed Page Tables with an example.	CO3	Ap	5
5.	a)	Consider the following access matrix	CO3	Ev	5

domain \ object	F_1	F_2	F_3	printer
D_1	read		read	
D_2				print
D_3		read	execute	
D_4	read write		read write	

		Now, how can you implement copy rights, owner rights and control rights in the given access matrix?			
5.	b)	Suppose you have two parties, Alice and Bob. Alice wants to send an encrypted message to Bob. They both have their key pairs. Alice's public key (e, n): (7, 77) Alice's private key (d, n): (43, 77) Alice wants to send the message "M" to Bob, but she doesn't want anyone else to read it. The message "M" is represented as a numerical value: M = 9. Using Alice's public and private keys mentioned above, calculate the encrypted value C of the message M = 9 following the RSA encryption process. Then, calculate the decrypted message "M" using Alice's private key.Provide the numerical values for C and M after the encryption and decryption steps.	CO3	Un	5
		OR			
5.	b)	Trojan horse, Spyware, Ransomware and Trap door are the popular malware to breach the security of a system. Create a security plan to protect against these threats.	CO3	Un	5

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Final Examination, Autumn 2023

Course Code: CSE 3635 Course Title: Artificial Intelligence

Total marks: 50

Time: 2 hours 30 minutes

The figures in the right-hand margin indicate full marks.

Course Outcomes and Bloom's Levels are mentioned in an additional Column

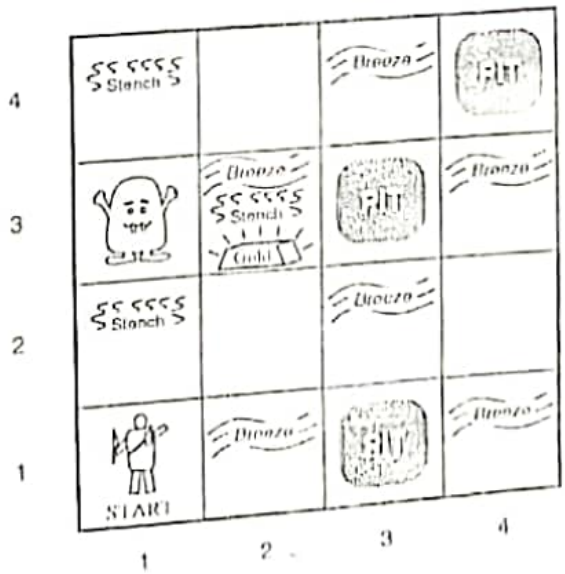
Course Outcomes (COs) of the Questions	
CO1	Understand the modern view of AI as the study of agents that receive precepts from the environment and perform actions.
CO2	Demonstrate awareness of informed search and exploration methods.
CO3	Apply AI techniques for knowledge representation, planning and uncertainty management.
CO4	Analyze different decision making and learning methods.

Bloom's Levels of the Questions						
Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analysis	Evaluate	Create

Part A

[Answer the following questions]

			CLO	BL	
1.	a)	What is knowledge-base? Draw and discuss the architecture of a knowledge-based agent.	CO3	U	1+2
	b)	The figure below shows a Wumpus World game, where the agent starts from location [1,1]. Briefly discuss with diagrams how the agent generates new knowledge through perceiving the environment and choosing his next move.	CO3	Ap	5



c) State the ontological and epistemological commitment of the propositional, first-order, probability theory, and fuzzy logic. — CO3 | U | 2

OR,

1\. a) What is knowledge-base? Write the tasks of TELL and ASK function in a knowledge based system. — CO3 | U | 3

b) In a knowledge-based system, the following rules are given: — CO3 | Ap | 5

Rule 1: A breeze in cell B22 implies that there is a pit in either cell P21, P23, P12, or P32.

Rule 2: There is no breeze in cell B22.

Using logical rules such as biconditional elimination, modus ponens, disjunction elimination, and other necessary logical rules, determine

$\neg P_{21} \wedge \neg P_{23} \wedge \neg P_{12} \wedge \neg P_{32}$.

c) How would you model the speed of a vehicle (65km/h) in both fuzzy and crisp sets? — CO3 | U | 2

2\. a) Differentiate between forward chaining and backward chaining. In what aspects forward chaining should be used? — CO3 | U | 2+2

b) Write down the top-down and bottom-up parse tree for the sentence "Book the flight through Houston" using the following CFG. — CO4 | Ap | 3

• Grammar	Lexicon
S → NP VP	Det → the | a | that | this
S → Aux NP VP	Noun → book | flight | meal | money
S → VP	Verb → book | include | prefer
NP → Pronoun	Pronoun → I | he | she | me
NP → Proper-Noun	Proper-Noun → Houston | NWA
NP → Det Nominal	Aux → does
Nominal → Noun	Prep → from | to | on | near | through
Nominal → Nominal Noun	
Nominal → Nominal PP	
VP → Verb	
VP → Verb NP	
VP → VP PP	
PP → Prep NP	

c) Explain with example why universal quantification elimination is easy while existential quantification elimination is not that much easy. — CO3 | U | 3

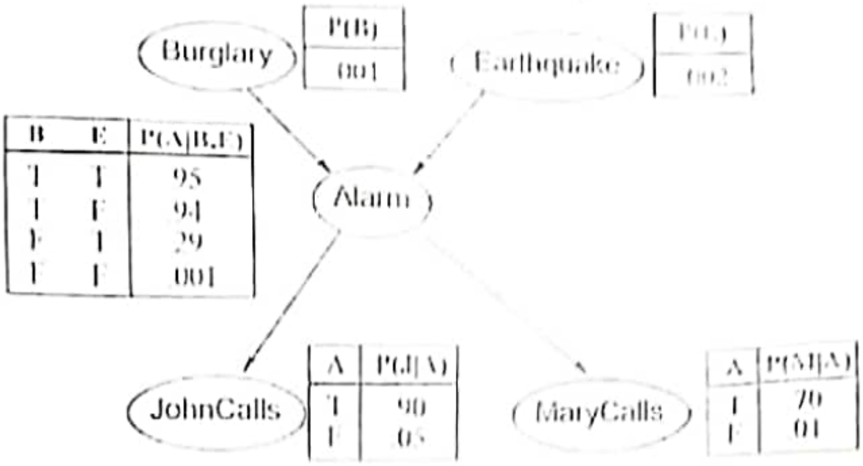
Part B

[Answer the following questions]

3\. a) Explain the Bayes Theorem. — CO4 | U | 2

3\. b) A doctor knows that the disease meningitis causes the patient to have a stiff neck, say, 40% of the time. The doctor also knows some unconditional facts: the prior probability that a patient has meningitis is 1/50000, and the prior probability that any patient has a stiff neck is 1/25. Find the probability of patients with a stiff neck having meningitis. — CO4 | Ap | 3

3\. c) A Bayesian network (Figure 1), showing both the topology and the conditional probability tables (CPTs). In the CPTs, the letters B, E, A, J, and M stand for Burglary, Earthquake, Alarm John Calls, and MaryCalls, — CO4 | Ap | 5 respectively. The Independent conditional probability helps us to write in a simplified way the joint distribution P(B, E, A, J, M).



i) Express the joint distribution P(B, E, A, J, M) in terms of the conditional probabilities (and independencies) expressed in the Bayesian Network above.

ii) Probability of the event that the alarm has sounded but neither a burglary nor an earthquake has occurred and both Mary and Jhon call.

iii) Probability of the event that the alarm has sounded and Burglary has occurred, an earthquake has not occurred, and both Mary and Jhon call.

4. a) The following probability distributions are given: CO3 Ap 5

$P(W)$

R	P
sun	0.8
rain	0.2

$P(D|W)$

D	W	P
wet	sun	0.1
dry	sun	0.9
wet	rain	0.7
dry	rain	0.3

Using Bayes theorem find P(W | dry) ?

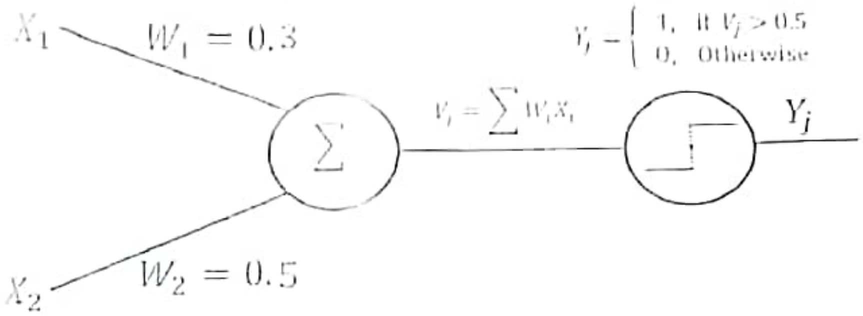
b) Suppose the reputation of a university depends on it's academic excellence and infrastructure. These attributes are graded as poor, average, or good, with specific weights and degrees of belief assigned to each evaluation grade for assessment. CO3 Ap 5

Evaluation Grade H_1=poor; H_2= average, H_3= good	Weight	Belief		
	Wi	β_{1j}	β_{2j}	β_{3j}
Accuracy	0.35	0.4	0.5	0
Reputation	0.65	0.1	0.75	0.15

Calculate the remaining belief and probability mass of the following attributes using DS theory.

Evaluation Grade H_1=poor, H_2= average, H_3= good	Belief	Probability mass					
	β_H	$m_{1,i}$	m_2	$m_{\theta,i}$	$m_{H,i}$	$\bar{m}_{H,i}$	$\tilde{m}_{H,i}$
Accuracy							
Reputation							

5. a) Consider the following neural network model with a learning rate of 0.2. CO AP 4



Demonstrate the learning process of the OR operation as shown below using the delta rule.

b) How do single-layer recurrent neural networks differ from multilayer feedforward neural networks? CO U 3

c) What is natural language processing (NLP)? Briefly discuss the steps through which NLP is conducted. CO U 3

OR,

5. a) What is natural language processing (NLP)? Describe the different phases of natural language processing (NLP) with example. CO U 4

b) "*Fuzzy set is the generalized version of crisp set*"- Do you agree with the statement? Justify your answer. CO AN 3

c) What do you mean by uncertainty? What are the different sources of uncertainty. CO U 3

International Islamic University Chittagong

Dept. of Computer Science & Engineering

Course Code: 3640, Course Title: Software Development 2 Sessional

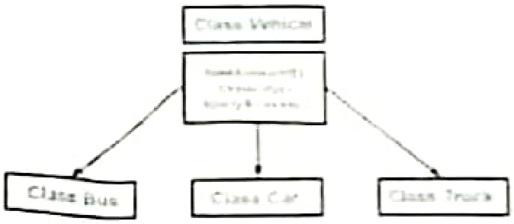
Time: 3 hour

Final Examination, Autumn-2023

Marks: 50

1) **Quiz :** 10

1. Which OOP's feature allows showing important aspects while hiding implementation details?



2. Above class structure belongs to which terms of object oriented programming?
3. Define constructor.
4. ______________refers to implementing operators using user-defined types based on the arguments passed along with it.
5. Write the types of feasibility study.
6. Cohesion is a measure that defines the __________among the elements of the module.
7. How you can make sure that your written code which can handle various kinds of error situation?
8. Write the stages of SDLC.
9. Why try/catch block used for?
10. __________is used when there is a need for override functionality when inheriting class.

2) Project.

) Presentation.

Viva-voce

Dept. of CSE
Faculty of Science and Engineering
International Islamic University Chittagong

Student ID					Student Name	
Obtained Marks	Q. No	COs	Marks	Total	Course Title	Operating System Sessional
	1				Course Code	CSE-3632
	2				Term	Autumn-2023
	3				Exam	SEE
	4				Full Marks	50 marks
	5				Time	3 hour
	6				Date	1/12/2023

1. Write a program to solve the following problem. 20 marks

Five batch jobs, A through E, arrive at a computer center at almost the same time. They have estimated running times of 10, 6, 2, 4, and 8 minutes. Their (externally determined) priorities are 3, 5, 2, 1, and 4, respectively, with 5 being the highest priority. For each of the following scheduling algorithms, determine the mean process turnaround time. Ignore process switching overhead.

(a) Priority scheduling.
(b) Shortest job first.
Assume that only one job at a time runs, until it finishes. All jobs are completely CPU bound.

OR

Write a program to implement the deadlock detection algorithm with multiple resources of each type. Your program should read from a file the following inputs: the number of processes, the number of resource types, the number of resources of each type in existence (vector E), the current allocation matrix C (first row, followed by the second row, and so on), the request matrix R (first row, followed by the second row, and so on). The output of your program should indicate whether there is a deadlock in the system. In case there is, the program should print out the identities of all processes that are deadlocked.

2. **Project** 20 marks
3. **Quiz** 20 marks

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE

Final Exam, Autumn 2023

Course Code: **CSE 3525**

Course Title: **Data Communication**

Time: 2 hours 30 minutes

Full Marks: 50

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

Course Outcomes (COs) of the Questions	
CO1	Understand Data Communications Concepts and its components.
CO2	Analyze the different types of Transmission media and their functions within a Network.
CO3	Apply the knowledge of encoding, decoding, and how error correction and error detection in data communication.
CO4	Understand switching principles and basics of wireless communication.

Bloom's Levels of the Questions						
Letter Symbols	R	U	App	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Group A

[Answer the questions from the followings]

1.	a)	What do you mean by signal rate and data rate? Write the relationship between signal rate and data rate. An analog signal has a bit rate of 16000 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements do we need?	CO3	An	5
	b)	Create a constellation diagram for 8-QAM with 4 phases and two amplitudes. Also, provide the time-domain plot for the given digital information signal: 111000101101110001.	CO3	Ap	5
2	a)	A corporation has a 10 MHz satellite channel and aims to create 40 distinct, independent channels, each capable of transmitting at least 10 mbps. Design an appropriate multiplexing technique for this communication requirement.	CO3	Ap	5
	b)	Four sources, each with a bit rate of 1000 mbps, need to be combined using Time Division Multiplexing (TDM) with a byte interval and synchronizing bits. Answer the following questions regarding the multiplexing process: i. What is the size of a frame in bits? ii. What is the frame rate? iii. What is the duration of a frame? iv. What is the data rate?	CO3	Ap	5

OR,

2.	a)	Describe multiplexing with necessary example and figure. Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 40kHz between the channels to prevent interference?	CO1	U	5
	b)	Distinguish between synchronous and statistical TDM. You need to use synchronous TDM and combine 20 digital sources, each of 100 Kbps. Each output slot carries 1 bit from each digital source, but one extra bit is added to each frame for synchronization. Answer the following questions: i. What is the size of an output frame in bits? ii. What is the output frame rate? iii. What is the duration of an output frame? iv. What is the output data rate?	CO4	U	5

Group B

[Answer the questions from the followings]

3.	a)	What is virtual circuit switching? Describe the virtual circuit setup, data transfer, and teardown phases with illustrations.	CO4	U	5
	b)	Compare the delay in Circuit-Switched and Packet-Switched Networks using appropriate examples.	CO4	U	5
4.	a)	A sender needs to send the four data items Ox3456, OxABCC, Ox02BC, and OxEEEE. Answer the following. i. Find the checksum at the receiver site if the second data item is changed to OxABCE. ii. Find the checksum at the receiver site if the second data item is changed to OxABCE and the third data item is changed to Ox02BA.	CO1	U	5
	b)	Discuss the concept of redundancy in error detection and correction. Given the dataword 1010011110 and the divisor 10111. i. Show the generation of the codeword at the sender site (using binary division). ii. Show the checking of the codeword at the receiver site (assume no error).	CO3	App	5
5.	a)	Compare and contrast the Stop-And-Wait ARQ with the Go-Back-N ARQ flow control protocol.	CO2	An	5
	b)	Explain, with appropriate diagrams, the consequences of not using an adequate window size for the Selective-Repeat ARQ flow control protocol.	CO2	An	5

OR,

5.	a)	What do you mean by SONET? Find the data rate of an STS-9 signal used in SONET.	CO3	App	5
	b)	How does the Ethernet address 1A:2B:3CAD:5E:6F appear on the line in binary? Compare a piconet and a scatternet.	CO1	An	

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International Islamic University Chittagong

Morality Development Program (MDP)

Semester Final Examination, Autumn-2023

Faculty of Science and Engineering, Semester- 6th

Course Title: Islamization of Discipline — Course Code: MDP-3606

Time: 2:30 Hours — Total Marks-50

Answer the following questions; Figures in the right hand margin indicate full marks

1. a) What do you mean by Finger Print of human being — 2
   b) How pain receptors work in the human body ? — 3
   c) What are the Ethical issues in Computer Use? Write short notes on it from Islamic Perspective — 5

2. a) How many are the pillars of Islam, and what are they ? — 2
   b) Write down the effects of salah on human personal life and social life from the perspective of the Qur'an and sunnah — 5
   c) What is autophagy ? How autophagy works in the human body — 3

3. a) "Algebra" is named after which book? Who is the author of the book ? — 2.5
   b) Who is called the doctor of all doctors? Who is the father of Chemistry — 2.5
   c) Name two Muslim scholars who had the contributions in Mathematics and Geography — 2.5
   d) Write down the name of two scientists who contributed in the field of Medicine — 2.5

4. a) Where the Lot Sea is located? What was the crime of Lot's tribe? — 2
   b) What are the negative impacts of having sex outside of marriage, homosexuality, pornography and short dress on society ? — 6
   c) Mention two examples of natural disasters described in the glorious Qur'an — 2

   Or

   a) What is Miracle (Mojeza)? Mention four different miracles by four different messengers by mentioning Quranic verses. — 5
   b) "The Qur'an itself is a miracle "- explain this statement with your logical argument — 5

5. What were the various approaches and steps taken by Bangabandhu to protect Islamic values, create religious awareness and establish peace and order in the country ? — 10

   Or

   a) Write down the flow chart of educational psychology — 3
   b) Discuss how the learning process works. — 3
   c) Describe the traits of an ideal student — 4

International Islamic University Chittagong
Department of Computer Science & Engineering
B.Sc. in CSE Semester Final Examination, Autumn 2023
Course Code: CSE-3637 Course Title: Software Engineering
Time: 2 hours 30 minutes Total Marks: 50

[Answer all the questions. Use separate answer script for Group A and Group B. Figures on the right margin indicate marks.]

Group A

1. Read the passage and answer questions 1(a-b):
In the bank's automated teller machine(ATM) system, customers interact directly with the bank's account database. In addition, there are bank management and security personnel who do not directly use the bank's ATM system but influence the overall system by handling all operations. And finally, there have been some standards developed for interbank communications.

a) What is the viewpoint? Write the name of all types of viewpoints. 3 CO2

b) Find out all types of viewpoints on the bank's ATM system and draw a hierarchy. 7 CO2

2.a) Write the difference between Coupling and Cohesion. 2 CO

2.b) What are the different activities performed by a User Interface Design? 5 CO

2.c) Imagine you are the lead software architect for a project to develop a Bus seat booking system for IIUC Transportation. Explain the concept of design quality in the context of software development and discuss why it's critical for this type of application. 3 CO

Or

2.a) Imagine you are tasked with designing software "Research work Tracker" for IIUC. Describe the key design considerations and architectural decisions you would make for this system. 3 CO2

2.b) *Explain* the concept of domain-specific architecture (DSA) in software design. Discuss the advantages of using domain-specific architectures. 3 CO

2.c) *Differentiate* between object-oriented design (OOD) and component-level design (CLD) in software development. Explain the core principles of each approach and provide examples of situations where OOD and CLD are most suitable. 4 CC

Group B

3 Read the passage and answer questions 3(a-b):
In a software development company, a team of skilled testers was responsible for ensuring the products' quality. They utilized various testing methods, such as black-box testing and white-box testing.

One day, the team was assigned a new project involving the creation of a mobile application for a social networking platform. As the project progressed, the testers were tasked with thoroughly testing the application before its release to identify any bugs or issues.
Now, imagine that you are a member of this team's testing department. Using your knowledge of black box and white box testing, please answer the following scenario-based question.

You decide to employ both black box and white box testing techniques for the social networking application as part of your testing strategy. The black box testing methodology entails testing an application from an external perspective, without knowing its internal structure or source code. In contrast, white box testing requires an examination of the application's internal workings, including the code, algorithms, and data structures

a) What specific scenarios or features would you focus on testing using black box techniques, and what scenarios or areas would you prioritize for white box testing? 6 C

b) How would you choice of testing techniques contribute to ensuring the overall quality and functionality of the social networking application? 4 C

4.a) What is software maintenance? Explain the types of software maintenance. 3

4.b) Elaborate on the process of software re-engineering and discuss the benefits of software re-engineering. 3

4.c) What is legacy system? Discuss the layers of legacy system. 3

4.d) What is 'Bad smells' in program code? 1

5.a) Suppose the fan out of a module is: 4, the cumulative number of variables passed to and from the module is: 47, and the complexity of the module is: 40. Use Card and Glass's Systems Complexity in order to find out the structural complexity of the module. 4

b) A simple stand-alone software utility is to be developed in 'C' programming by a team of software experts for a computer running Linux. The overall size of this software is estimated to be 20,000 lines of code. Considering (a, b) = (2.4, 1.05) as multiplicative and the exponential factor for the basic COCOMO effort estimation equation and (c, d) = (2.5, 0.38) as multiplicative and exponentiation factor for the basic COCOMO development time estimation equation, approximately how long the software project takes to complete? 4

c) Write short note on: 2

A. Legacy System
B. COCOMO Models
C. Requirement Engineering Process
D. Henry and Kafura's metric

Or

5.a) Explain the significance of 'Fan in' and 'Fan out'. How do they contribute to measuring the complexity of a software system? Provide an example of a situation where 'Fan in' and 'Fan out' metrics could be used to assess software complexity. 3

5.b) Discuss the role of Henry and Kafura's metric, Card and Glass's Systems Complexity metric. 3

5.c) Explain the concept of software reliability and its significance in software engineering. Discuss at least three software reliability metrics that are commonly used to assess the reliability of a software system. 4